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1. Domain Archaea

This domain includes single celled organisms, the prokaryotes which have the ability to grow in extreme conditions like volcano vents, hot springs and polar ice caps, hence are also called **extremophiles.** They are capable of synthesizing their food without sunlight and oxygen by utilizing hydrogen sulphide and other chemicals from the volcanic vents. Some of the them produced methane (methanogens), few live in salty environments (Halophiles) and are thermoacidophiles which thrive in acidic environments and at high temperatures.

2. Domain Bacteria

Bacteria are prokaryotic, their cells have no definite nucleus and DNA exists as a circular chromosomes and do not have histones associated with it. They do not possess membrane bound organelles except for ribosome (70S type). Their cell wall contains peptidoglycans. Many are decomposers, some are photo-synthesizers and few cause diseases. There are beneficial **probiotic** bacteria and harmful **pathogenic** bacteria which are diversely populated. Cyanobacteria are photosynthetic blue green algae which produce oxygen. These had played a key role in the changes of atmospheric oxygen levels from anaerobic to aerobic during the early geologic periods.

Curd is one of the best sources of probiotics, which are friendly bacteria that can improve our health. e.g. *Lactobacillus sp.*

3. Domain Eukarya (Eukaryotes)

Eukaryotes are animals which have true nucleus and membrane bound organelles. DNA in the nucleus is arranged as a linear chromosome with histone proteins, ribsosomes of 80S type in the cytosol and 70S type in the chloroplast and mitochondria. Animals in this domain are classified under kingdoms, namely, Protista, Fungi, Plantae and Animalia.

In 1987, **Cavalier-Smith** revised the six kingdom system to **Seven Kingdom system.** The concept of super kingdom was introduced and revised to seven kingdom classification. The classification is divided

FIVE KINGDOM CLASSIFICATION

Salient features	KINDS OF KINGDOM				
	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Non-cellular	Present in some	Present	Present	Absent
Body organisation	Cellular	Cellular	Multicelluar Tissue	Tissue Organ	Tissue Organ Organ system
Mode of nutrition	Autotrophic Heterotrophic	Autotrophic Heterotrophic	Heterotrophic	Autotrophic	Heterotrophic

Table 1. Five Kingdom Classification